FAIRagro

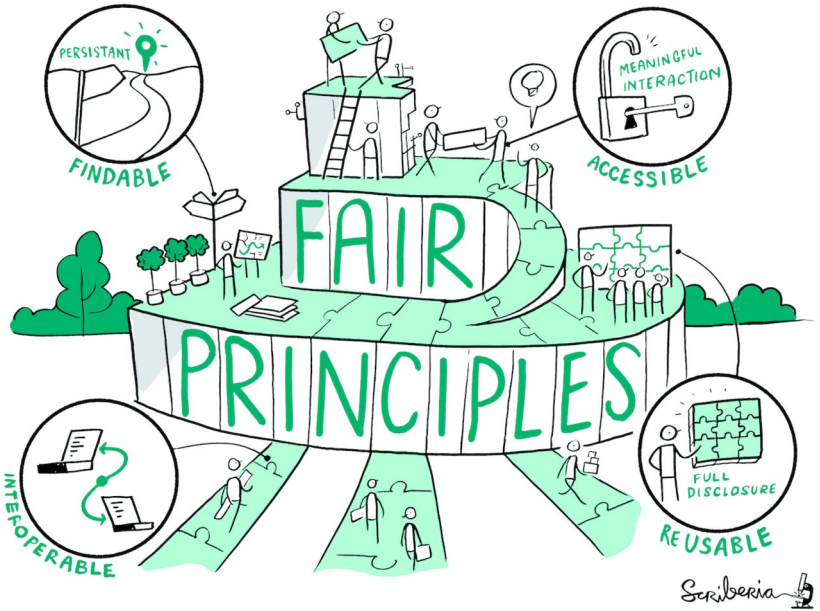
# FAIR Principles

# What are the FAIR data principles?

Set of principles that **may** act as a **guideline** for those wishing to enhance the reusability of their data holdings.

**FAIR** stands for:

- **Findability**
- **Accessibility**
- **Interoperability**
- **Reusability**




Source: Wilkinson et al. 2016 modified by ZB MED 2022, Illustration: Image by The Turing Way Community & Scriberia (2023) is licensed under CC BY 4.0

# What makes your data Findable?

- Assign a globally unique and **persistent identifier** (e.g. DOI) to your (meta)data
- **Describe** your data with rich metadata
- Metadata include the **identifier** of the data it describes
- Register the (meta)data in a **searchable resource** (e.g. data repository)




Source: Modified from Wilkinson et al. 2016 by Vandendorpe & Lindstädt 2020, Image by The Turing Way Community & Scriberia (2023) is licensed under CC BY 4.0

# What makes your data Accessible?

- Access without special or proprietary tools
- Use a **standardized communications protocol** for your (meta)data:
  - The protocol is **open**(-sourced), **free** (no-cost), and **universally implementable** (e.g. HTTP, FTP, …)
  - The protocol allows for an **authentication** and **authorization procedure**, where necessary
- **Metadata** are **accessible**, even when the data are no longer available




Source: Modified from Wilkinson et al. 2016 GO FAIR by Vandendorpe & Lindstädt 2020 and Rathmann et al. 2021,
Image by The Turing Way Community & Scriberia (2023) is licensed under CC BY 4.0

# What makes your data Interoperable*?

- Use for your (meta)data:
  - **vocabularies** & **thesauri**
  - **language for knowledge representation** (ontologies)
- Include in your (meta)data **qualified references** to other (meta)data




*** Data interoperability** is the ability of a dataset to work with other datasets or systems without special effort on the part of the user.

Source: Modified from Wilkinson et al. 2016, GO FAIR and Open Data MOOC by ZB MED 2022, Image by The Turing Way Community & Scriberia (2023) is licensed under CC BY 4.0

# What makes your data Reusable?

- Describe your (meta)data with **accurate** & **relevant** attributes
- Release your (meta)data with a clear and accessible **data usage license**
- Associate your (meta)data with **detailed provenance**




**A good example:**

Papoutsoglou, E. A. (2022) „A case-study for improved reusability of plant phenotyping data with MIAPPE“. Zenodo. doi: 10.5281/zenodo.5572772

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# FAIR Data = Open Data?

Data may be consistent with FAIR principles, but may have limited access.

- If you have sensitive data, you have to think about:
  - Ethical, legal and technical issues
  - Pay special attention to data collection, processing, handling, & storage

**As open as possible, as closed as necessary**

Source: Modified from de.NBi (2025)

# Exercise: FAIR or not FAIR?

**Case 1:** The results of a study for which interviews were conducted among farmers on their pest management interactions has been published in a professional journal. The associated data are located on a relevant repository and are appropriately tagged with metadata and linked to the article.

One of the farmers withdraws their consent for the publication of the data for personal reasons. Therefore, all data are deleted from the repository, but the metadata remain and are supplemented by an explaining note.

Source: Adapted from Bobrov et al. 2021 by FAIRagro

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→ **Solution:** FAIR

Source: Adapted from Bobrov et al. 2021 by FAIRagro

# Exercise: FAIR or not FAIR?

**Case 2:** For a master's thesis in agricultural science, soil samples are analyzed by a machine. The student compiles extensive metadata for the publication of the data. To be sure that the correct version of the experimental protocol of the analyzing machine is used, they take a photo of the type plate and provide it under the name "protocol used".

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→ **Solution:** not very FAIR

Source: Adapted from Bobrov et al. 2021 by FAIRagro

# Exercise: FAIR or not FAIR?

**Case 3:** An international group of several universities is working closely with a company from the agricultural engineering industry to develop a new field robot. The findings from this collaboration are presented in overview articles and the data are stored and made accessible in a repository set up specifically for this project. The data is only accessible to people within the group, as several patent cases are pending. Externals can only access the metadata referencing this item.

Source: Adapted from Bres et al. 2022 by FAIRagro

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**Case 3:** An international group of several universities is working closely with a company from the agricultural engineering industry to develop a new field robot. The findings from this collaboration are presented in overview articles and the data are **stored** and made **accessible** in a **repository** set up specifically for this project. The data is only accessible to people within the group, as several patent cases are pending. Externals can only **access the metadata** referencing this item.

→ **Solution:** FAIR

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FAIRagro

# Policies and Guidelines

# Definition of Research Data Policies

"Guidelines which apply to all members of an institution (e.g. a university) and govern how research data are to be managed." [Forschungsdaten.info 2024]

- ➔ institutional policies are **legally binding** for their employees
- ➔ not just at academic institutions, but also **funding organisations** and **journals**
- ➔ **can be the legal basis for a scientists right to be cited**



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# Definition of Research Data Policies

- can be discipline specific
- can be implemented on institutional level
- can be implemented for specific research project
- can be implemented by journals



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# The Guidelines for Good Research Practice

DFG Guidelines for Safeguarding Good Research Practice = overarching requirements that apply to research work for safeguarding good research practice.



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from Noun Project

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# The Guidelines for Good Research Practice

## 11 out of 19 Guidelines are RDM-relevant!

- Guideline 7: quality assurance across phases
- Guideline 8: Stakeholders, responsibilities and roles
- Guideline 9: Research design
- Guideline 10: legal and ethical frameworks, rights of use
- Guideline 11: methods and standards
- Guideline 12: documentation
- Guideline 13: establishing public access to research results
- Guideline 14: authorship
- Guideline 15: publication organ
- Guideline 17: archiving
- Guideline 19: Procedures in cases of alleged research misconduct




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Statements regarding using rights and citing

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**Worth reading: https://wissenschaftliche-integritaet.de/en/**




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